Solution: Tutorial Sheet-03

1. **Existence of Moments**
   Given PDF: $f(x) = \frac{2}{x^3}$ for $x \geq 1$ and 0 otherwise.

   (a) **Expectation $E(X)$:**

   $$E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_{1}^{\infty} x \cdot \frac{2}{x^3}dx = \int_{1}^{\infty} \frac{2}{x^2}dx.$$
   $$= \left[-\frac{2}{x}\right]_1^{\infty} = 0 - (-2) = 2.$$

   Since the integral converges to a finite value, $E(X)$ exists and equals **2**.

   (b) **Second Moment $E(X^2)$:**

   $$E(X^2) = \int_{1}^{\infty} x^2 \cdot \frac{2}{x^3}dx = \int_{1}^{\infty} \frac{2}{x}dx.$$
   $$= 2[\ln x]_1^{\infty} = \infty.$$

   The integral diverges, so $E(X^2)$ **does not exist**. **Implication:** Since the variance is $Var(X) = E(X^2) - (E(X))^2$, if $E(X^2)$ is infinite, the variance does not exist (is infinite). Furthermore, if a moment of order k does not exist, no moments of order $n > k$ exist. Thus, no higher-order moments exist.

2. **Discrete Expectation (Convergence)**
   Given PMF: $P(X = k) = \frac{6}{\pi^2 k^2}$ for $k = 1, 2, \ldots$

   $$E(X) = \sum_{k=1}^{\infty} k \cdot P(X = k) = \sum_{k=1}^{\infty} k \cdot \frac{6}{\pi^2 k^2} = \frac{6}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k}.$$

   The series $\sum \frac{1}{k}$ is the Harmonic series, which diverges to infinity. Thus, $E(X)$ **does not exist**.

   Justification for Absolute Convergence: For expectation to exist, we require $E[|X|] < \infty$. Consider a random variable Y taking values $x_k = (-1)^{k+1}\frac{2^k}{k}$ with probability $p_k = \frac{1}{2^k}$.

   - $\sum x_k p_k = \sum \frac{(-1)^{k+1}}{k}$ converges (Alternating Harmonic Series).
   - $\sum |x_k| p_k = \sum \frac{1}{k}$ diverges.

   In such a case, $E(X)$ is not well-defined (does not exist) because the sum depends on the order of summation.

3. **Expectation of a Function**
   Given $f(x) = 2x$ for $0 < x < 1$. Let $Y = e^X$. Using the Law of the Unconscious Statistician (LOTUS):

   $$E(Y) = E(e^X) = \int_{-\infty}^{\infty} e^x f(x)dx = \int_{0}^{1} e^x(2x)dx.$$

   Using Integration by Parts ($u = 2x, dv = e^x dx \implies du = 2dx, v = e^x$):

   $$= [2xe^x]_0^1 - \int_0^1 2e^x dx$$
   $$= (2(1)e^1 - 0) - [2e^x]_0^1 = 2e - (2e - 2) = \mathbf{2}.$$

4. **Symmetry**
   Given $f(x) = Ce^{-(x-2)^2}$.

   (a) **Center of Symmetry:** The exponent $(x-2)^2$ is symmetric about $x = 2$. Thus, $a = \mathbf{2}$.

   (b) **Values without Integration:**

   - Since the distribution is symmetric about 2 and the mean exists, $E(X) = \mathbf{2}$.
   - $E(X-2)^3$ is the third central moment. For any distribution symmetric about the mean, all odd central moments are zero. Thus, $E(X-2)^3 = \mathbf{0}$.

5. **Third Central Moment Relation**
   Let $\mu'_k = E(X^k)$ and $\mu = \mu'_1$.

$$\mu_3 = E[(X-\mu)^3] = E[X^3 - 3X^2\mu + 3X\mu^2 - \mu^3]$$

   Using linearity of expectation:

$$= E(X^3) - 3\mu E(X^2) + 3\mu^2 E(X) - \mu^3$$

   Substitute raw moments:

$$\begin{aligned} &= \mu'_3 - 3\mu'_1\mu'_2 + 3(\mu'_1)^2(\mu'_1) - (\mu'_1)^3 \\ &= \boldsymbol{\mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3}. \end{aligned}$$

6. **Variance Calculation**
   Given $E(X) = 2$ and $E(X^2) = 8$.

   (a) $Var(X) = E(X^2) - (E(X))^2 = 8 - (2)^2 = 8 - 4 = \mathbf{4}$.
   Standard Deviation $\sigma = \sqrt{4} = \mathbf{2}$.

   (b) $E(3X-5) = 3E(X) - 5 = 3(2) - 5 = \mathbf{1}$.
   $Var(3X-5) = 3^2 Var(X) = 9(4) = \mathbf{36}$.

7. **Degeneracy Variance**
   Proof:

   - ($\Leftarrow$) If X is degenerate at c, then $P(X = c) = 1$. Thus $E(X) = c$ and $E(X^2) = c^2$. $Var(X) = c^2 - c^2 = 0$.
   - ($\Rightarrow$) If $Var(X) = E[(X-\mu)^2] = 0$. Since $(X-\mu)^2 \geq 0$, for its expectation to be 0, the random variable must be 0 with probability 1. $P((X-\mu)^2 = 0) = 1 \implies P(X = \mu) = 1$. Thus X is degenerate.

8. **Median and Quantiles**
   Given CDF $F(x) = 1 - e^{-2x}$ for $x \geq 0$.

   (a) **Median (m):** Set $F(m) = 0.5$.

$$1 - e^{-2m} = 0.5 \implies e^{-2m} = 0.5 \implies -2m = \ln(0.5) = -\ln 2.$$

$$m = \frac{\ln 2}{2} \approx \mathbf{0.346}.$$

(b) **75th Percentile ($q_{0.75}$):** Set $F(q) = 0.75$.

$$1 - e^{-2q} = 0.75 \implies e^{-2q} = 0.25 \implies -2q = \ln(0.25) = -2\ln 2.$$

$$q = \ln 2 \approx \mathbf{0.693}.$$

9. **PGF**
$P(X = 1) = p, P(X = 0) = q$ where $q = 1 - p$.

   (a) $P(s) = E[s^X] = \sum s^k P(X = k) = s^0(q) + s^1(p) = \mathbf{q} + \mathbf{ps}$.

   (b) $P'(s) = p$. $E(X) = P'(1) = \mathbf{p}$.

10. **PGF**
Given $P(s) = e^{\lambda(s-1)}$.

   (a) Derivatives:

$$P'(s) = \frac{d}{ds}e^{\lambda s - \lambda} = \lambda e^{\lambda(s-1)}.$$

$$P''(s) = \frac{d}{ds}(\lambda e^{\lambda(s-1)}) = \lambda^2 e^{\lambda(s-1)}.$$

   (b) Moments:

$$E(X) = P'(1) = \lambda e^0 = \lambda.$$

$$E(X^2) - E(X) = P''(1) = \lambda^2. \implies E(X^2) = \lambda^2 + \lambda.$$

$$Var(X) = E(X^2) - (E(X))^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda.$$

11. **MGF Calculation**
$f(x) = e^{-x}$ for $x > 0$.

$$M_X(t) = E[e^{tX}] = \int_0^\infty e^{tx}e^{-x}dx = \int_0^\infty e^{-(1-t)x}dx.$$

$$= \left[\frac{e^{-(1-t)x}}{-(1-t)}\right]_0^\infty.$$

For the upper limit to converge to 0, we need $1 - t > 0 \implies t < 1$.

$$= 0 - \frac{1}{-(1-t)} = \frac{1}{1-t}, \quad \text{for } \mathbf{t < 1}.$$

Linear Transformation Property:

$$M_Y(t) = E[e^{tY}] = E[e^{t(aX+b)}] = E[e^{tb} \cdot e^{(at)X}] = e^{tb}E[e^{(at)X}] = e^{tb}M_X(at).$$

12. **Moments from MGF Expansion**
Given $M_X(t) = 1 + 3t + \frac{14t^2}{2!} + \dots$ The Maclaurin series is $M_X(t) = \sum \frac{E(X^n)}{n!}t^n$. Comparing coefficients:

   (a) Coeff of t: $E(X) = \mathbf{3}$.

   (b) Coeff of $\frac{t^2}{2!}$: $E(X^2) = \mathbf{14}$.

   (c) Variance: $Var(X) = 14 - 3^2 = 14 - 9 = \mathbf{5}$.

13. **MGF**
$f(x) = \frac{1}{2}e^{-|x|}$.

(a) $M_X(t) = \int_{-\infty}^{\infty} e^{tx}\frac{1}{2}e^{-|x|}dx$. Split the integral at 0:

$$= \frac{1}{2}\int_{-\infty}^{0} e^{tx}e^{x}dx + \frac{1}{2}\int_{0}^{\infty} e^{tx}e^{-x}dx$$

$$= \frac{1}{2}\int_{-\infty}^{0} e^{(1+t)x}dx + \frac{1}{2}\int_{0}^{\infty} e^{-(1-t)x}dx$$

- First integral converges if $1 + t > 0 \implies t > -1$. Value: $\frac{1}{1+t}$.
- Second integral converges if $1 - t > 0 \implies t < 1$. Value: $\frac{1}{1-t}$.

Combined Region: $\mathbf{-1 < t < 1}$.

$$M_X(t) = \frac{1}{2}\left(\frac{1}{1+t} + \frac{1}{1-t}\right) = \frac{1}{2}\frac{2}{1-t^2} = \frac{\mathbf{1}}{\mathbf{1 - t^2}}.$$

(b) The MGF exists only in the open interval $(-1, 1)$. It diverges for $|t| \geq 1$. However, moments of all orders exist because $E[X^n] = \int x^n \frac{1}{2}e^{-|x|}dx$ converges for any finite n (exponential decay dominates polynomial growth). This illustrates that existence of moments does not imply the existence of MGF for all real t.